Uncovering Cost Drivers and Fraud Patterns to Improve Claim Efficiency in U.S. Healthcare

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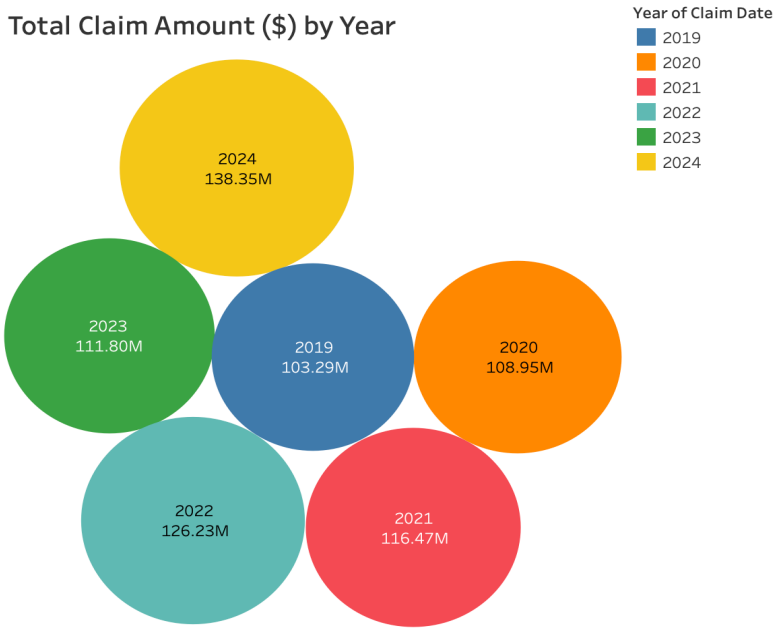
Rising U.S. healthcare costs highlight inefficiencies in claims. This analysis uncovers cost drivers, delay and fraud patterns, and high-impact segments to guide strategies that reduce financial leakage and improve claim efficiency.

Summary Table

Patient Gender	Claim Amount ($)	CoPay Amount	Deductible Amount	Total Claim
Female	386.59M	29,025,461	58.04M	11,052
Male	252.88M	18,907,626	37.93M	7,189
Other	65.61M	4,924,297	9.80M	1,859

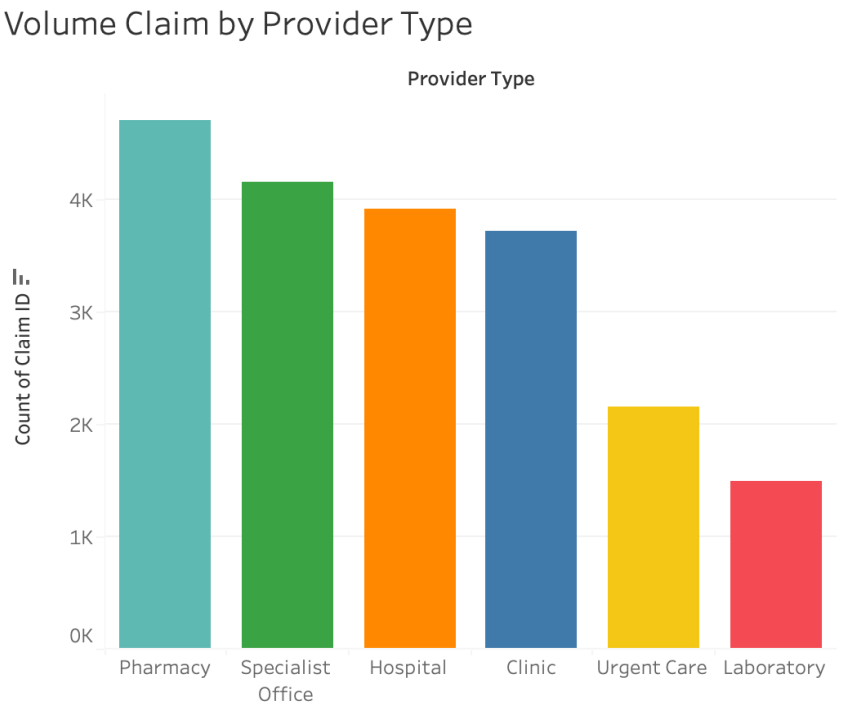


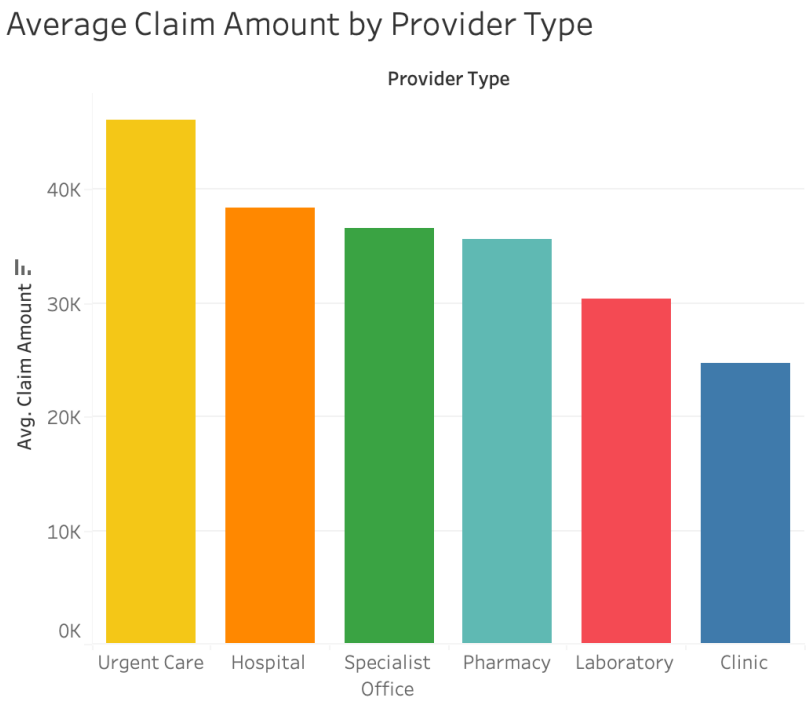
Executive Summary

- $7M+ annual fraud loss despite low claim volume (~7%).
- Urgent Care & Hospitals drive high costs; Pharmacy & Specialist Offices drive volume.
- Ages 40–60 and Trauma/Elective admissions are key cost hotspots.
- High-cost states: MN, TX, IL | High-volume states: NY, MN, IN.
- Late submissions stable (~3%) → ongoing inefficiencies.

Pharmacy ($386.6M) and Specialist Office ($329.5M) drive the highest claim volumes at moderate costs, while Urgent Care leads in average claim amount and Hospitals contribute significantly to total costs. Clinics and Laboratories have minimal financial impact.

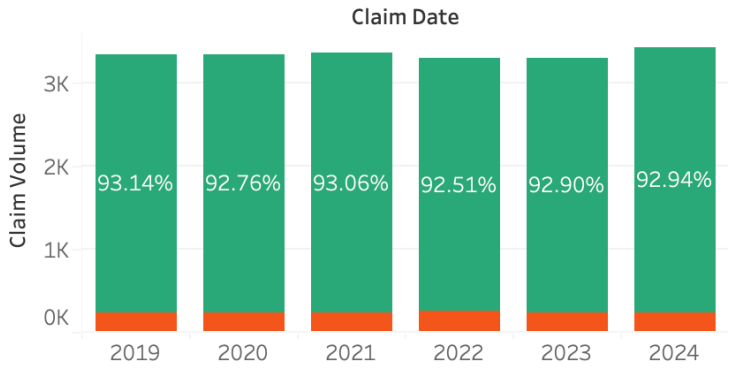
Provider Type
- Clinic
- Hospital
- Laboratory
- Pharmacy
- Specialist Office
- Urgent Care



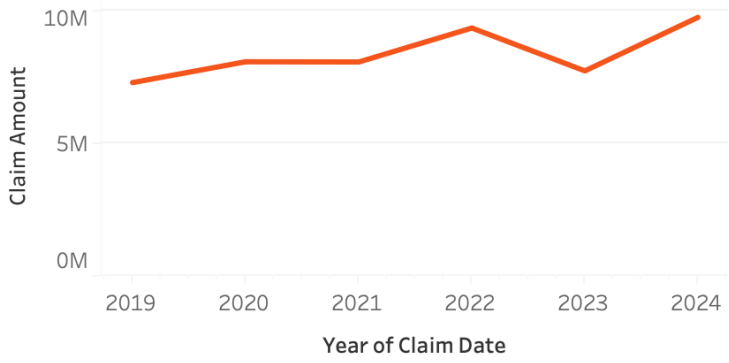


Fraudulent claims (~7%) drive over $7M in annual losses, peaking in 2022 and 2024, despite low volume. Late submissions remain stable at ~3%, indicating persistent inefficiencies. Strengthening fraud detection, claim validation, and submission timeliness is critical to reducing financial exposure.

Volume Fraud Claim by Year



Amount Loss/year for Fraud Claim



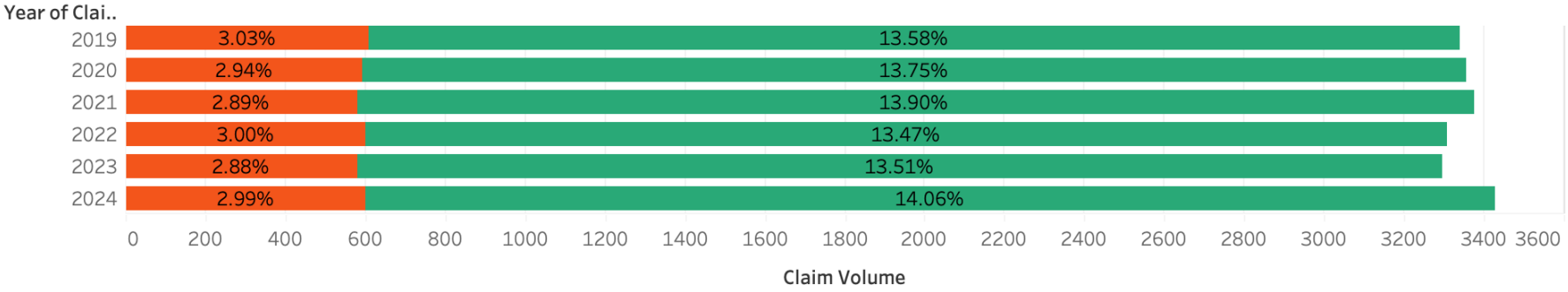
Is Fraudulent
- False
- True

Is Fraudulent
- (All)
- False
- ☑ True

Claim Submitted Late
- ☑ (All)
- ☑ False
- ☑ True

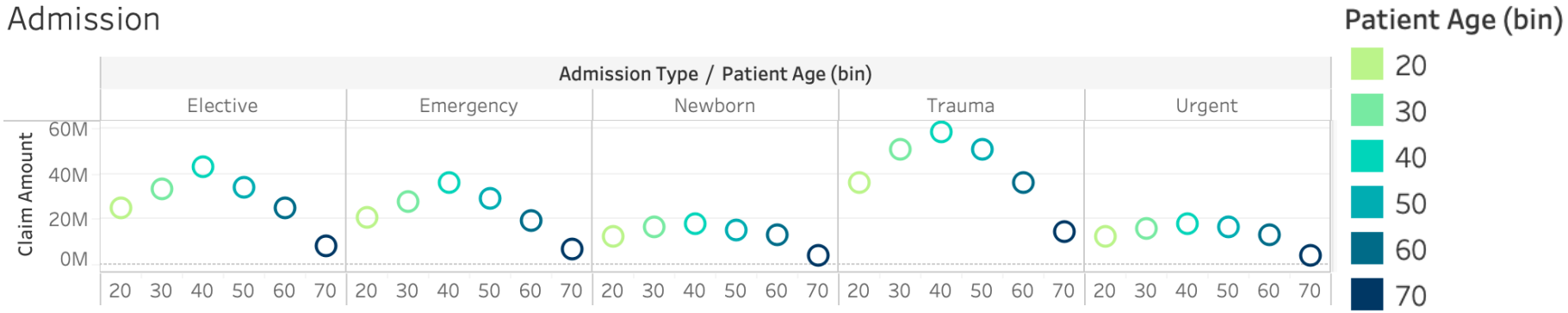
Claim Submitted Late
- False
- True

Volume Late Submission by Year

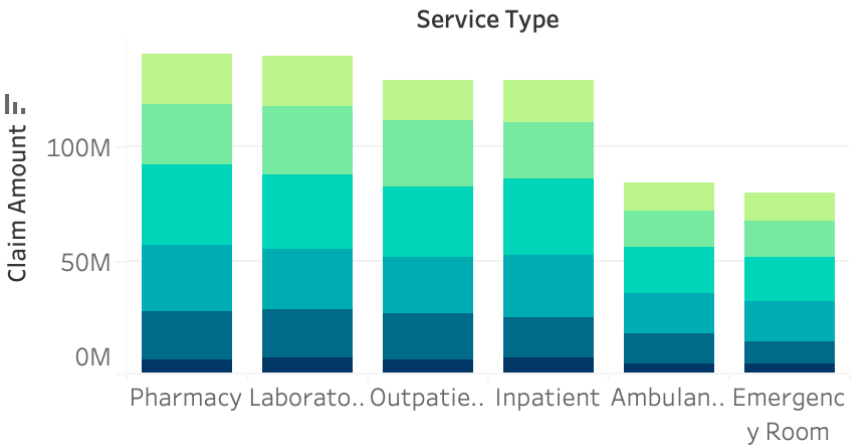


Trauma and Elective admissions drive the highest claim amounts, concentrated in ages 40–60. Pharmacy and Laboratory services lead service costs, while discharges "Against Medical Advice" and "Home" have the greatest financial impact, highlighting key cost hotspots across care pathways.

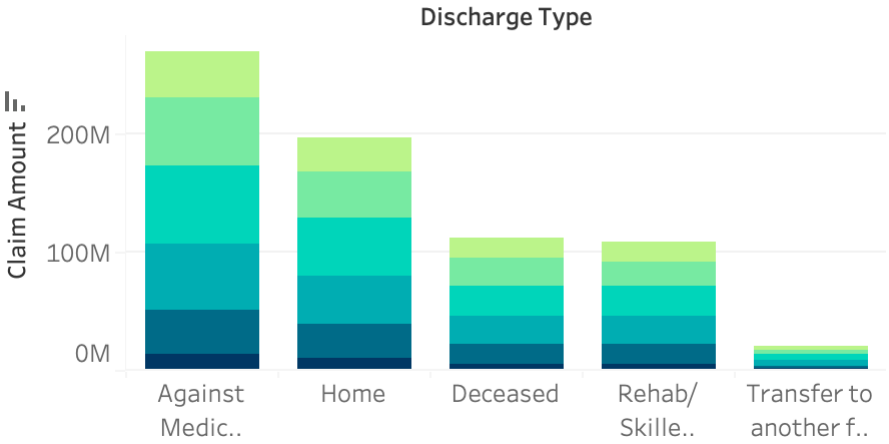
Admission



Service Type



Discharge Type

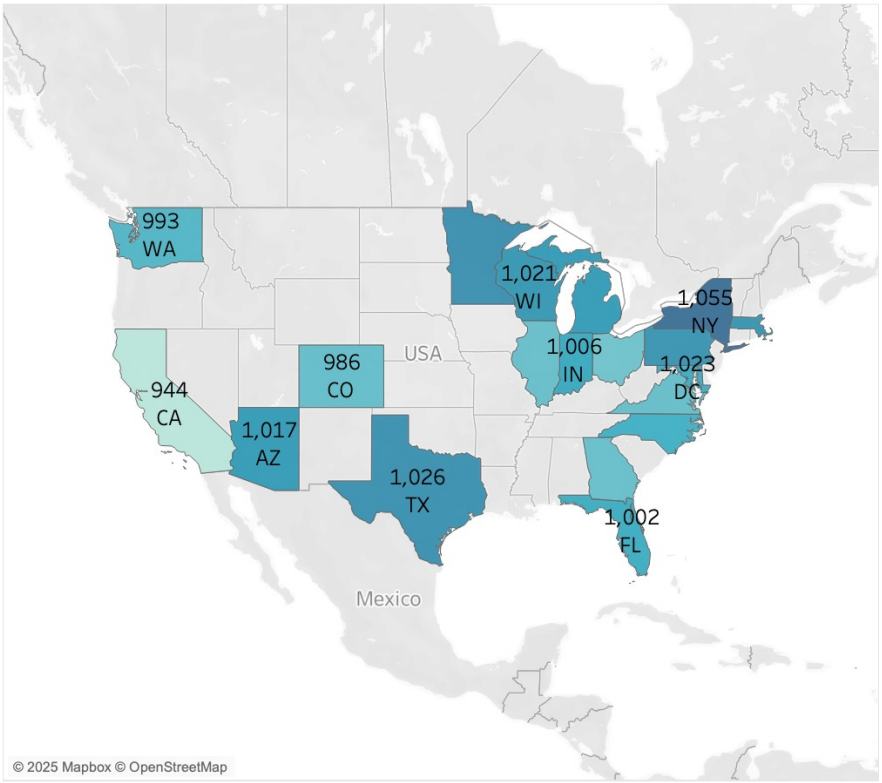


Minnesota, Texas, and Illinois record the highest average claim costs, while New York, Minnesota, and Indiana lead in claim volumes. Cost and volume patterns diverge, suggesting service mix drives costs more than volume, creating opportunities to reduce costs in high-cost states and optimize resources in high-volume regions.

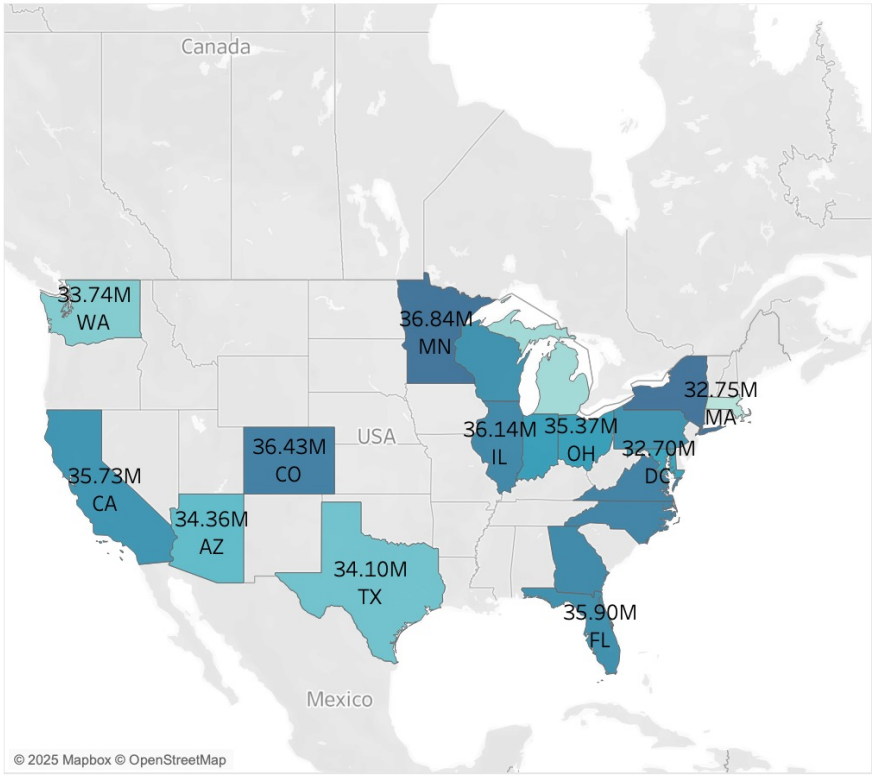
Claim Volume
944 1,055

Total Claim Amount ($)
32.70M 36.84M

Claim Volume by Location



Average Claim Cost by Location



Recommendations

Fraud Control: Strengthen detection and automate validation during peak years (2022 & 2024)

- Reduce annual fraud losses by 20–30% (~$1.4M–$2.1M)

Cost Optimization: Focus on Urgent Care & Hospital claims (35%+ of total cost)

- Achieve 5–10% reduction in cost per claim

Care Coordination: Improve follow-up for AMA and Home discharges

- Lower avoidable discharge-related costs by 10–15%

Geographic Strategy: Control costs in MN, TX, IL and optimize resources in NY, MN, IN

- Generate $3M–$4M in annual savings

Process Improvement: Tighten submission timelines and automate alerts

- Improve process efficiency by 15–20% and reduce late submissions (~3%)

Thank you